

Three-Channel Equivalence in the Unified Framework of Representation, Logic, and Intuition — Construction Earns Intuition

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Draft v0.1 (English) | 2026-08-11 Target: TMLR / COLM / Cognition (theory + empirical) **Framework position:** Part of the *Unified Framework of Representation, Logic, and Intuition* (manifesto: `Unified_Framework_Representation_Logic_Intuition.md`). This paper establishes the **construction→intuition** cognition: construction, intuition, and formal rewriting are three equivalent execution channels of one system; intuition is compiled construction. Companion: P1 (form→construction, representation + generalization theory), P2 (construction→intuition method, compilation). Evidence: exp50 (three-channel consistency), exp41 (symbol permutation), exp80 (step-skipping), exp53 (convergence). Single official run_exp judge.

Repository: [YuchenWang-ai/Three-Channel-Equivalence-Construction-Intuition-and-Formal-Rewriting-in-One-Neural-System](#)

Abstract

The century-old split in the foundations of mathematics — intuitionism (a mathematical object exists iff it can be constructed), structuralism (an object is its position in a structure), and formalism (mathematics is a sign game) — is mirrored in a single neural system as three execution channels for the same mathematical object. We show that, in a token-native representation, the *construction* channel (definition-driven unfolding), the *intuition* channel (compiled weight execution), and the *formal* channel (definition-driven rewriting) are pairwise semantically equivalent: on a 2000-digit carry chain all three produce identical results (1.000). We argue this is the neural implementation of Computational Trinitarianism (Curry–Howard: programs = proofs = types), with the crucial difference that the equivalence holds between *execution channels of one system* rather than between logics. We develop the epistemology: intuition is the reasoning fast path — hard to obtain (config-dependent epochs), but easy to generalize by skipping construction steps; symbol permutation invariance (0.987) and zero-decay step-skipping (2000 digits, radix 60, 100% anonymization) establish that the intuition channel carries structure, not symbol semantics. Stance: anti-Platonist — intuition is compiled construction, not innate. Within the Unified Framework, this paper is the **construction→intuition** cognition leg: correct construction yields intuition (compiled construction), and correct generalization compiles into flashing intuition (the fast reasoning path).

1 Introduction

The split. In the foundations of mathematics, three answers compete for the question “what is a mathematical object?” Brouwer’s intuitionism: an object exists iff it can be *constructed*. Bourbaki’s structuralism: an object is its *position in a structure*. Hilbert’s formalism: mathematics is a *sign game*; an object is its role in a rewriting system. The dispute is usually presented as philosophical; in this paper we observe that it is also *architectural* — the three positions correspond to three ways a computing system can realize the same object.

The unification. A single neural system over a token-native representation (P1) exhibits all three channels: - *Construction* (intuitionism): unfold the definition chain, step by step — existence by construction; - *Intuition* (structuralism): compiled weights position the object directly

— existence by location in the learned structure; - *Formal* (formalism): the definition-driven rewriting engine — existence by the sign-manipulation rules.

We show experimentally that the three channels are pairwise semantically equivalent on the same inputs (exp50: construction = formal exactly; intuition 1.000 on a 2000-digit carry chain). This is the neural implementation of Computational Trinitarianism — Curry-Howard’s programs = proofs = types — but with a decisive difference: the equivalence is *within one system’s execution channels*, not between distinct logical systems.

Why this matters. If construction, intuition, and formal rewriting agree on the same object, then the philosophical positions are not alternatives to be argued about but *facets of a single realization*. The disagreement over “what an object is” becomes a disagreement over *which channel to look through* — and all channels answer the same way. The engineering content is the compilation relation: construction can be compiled into intuition (P2), and both agree with rewriting.

Contributions.

- **C1 (Epistemology) — Intuition is compiled construction.** The intuition channel is not a third mysterious faculty; it is the construction process compiled into weights. Anti-Platonist: no innate structure, no a priori object.
- **C2 (Empirics) — Three-channel equivalence.** Construction = formal exactly, and intuition 1.000 on the same inputs, including a 2000-digit carry chain (exp50).
- **C3 (Empirics) — Intuition carries structure, not symbols.** Symbol permutation invariance (0.987, exp41) and zero-decay step-skipping (exp80) show the intuition channel realizes the structure of the objects, not their signifiers.
- **C4 (Epistemology) — Acquisition and step-skipping are decoupled.** Intuition is hard to obtain (config-dependent epochs; 3 simple / 8 full, exp53) but easy to generalize by skipping construction steps — the two facts are not in tension.

2 Three Positions, Three Channels

Position	Object is	Access	Our channel	Execution
Intuitionism (Brouwer)	a construction	construct it	Construction	unfold definition chain, step by step
Structuralism (Bourbaki)	a position	locate it	Intuition	compiled weights place it directly
Formalism (Hilbert)	a sign game	play it	Formal	definition-driven rewriting

The three channels share a single definition source: the token-native representation (P1 §4). Construction and formal read the same definitions; intuition is a compiled image of the same definitions. Equivalence is therefore not a coincidence of three independent systems but a property of one system viewed through three channels.

Computational Trinitarianism. Curry-Howard identifies programs, proofs, and types as the same thing under three presentations. Our claim is analogous: construction, intuition, and formal rewriting are the same computation under three presentations — *within one system*. The difference from Curry-Howard is the locus of equivalence: not “program = proof = type” as distinct entities, but “one object’s three execution paths agree.”

3 Intuition as Compiled Construction

The claim. The intuition channel is the construction process compiled into weights. Training (P1 §5) supervises the model with judgment outcomes while *excluding answer traces*: the model cannot copy the unfolding; it must induce the computation from definitions and compile it. The compiled product is intuition — the reasoning fast path.

Hard to obtain. Compilation is not free. Acquisition is config-dependent: 3 epochs for the simple suite, 8 for the full logic-and-arithmetic suite (exp53). Until compilation completes, the intuition channel does not generalize (exp53: epochs 1-3 at 0.1-24%). This is the *obtain* side of the decoupling.

Easy to generalize. Once compiled, the intuition channel generalizes by *skipping construction steps*: it jumps from input to output over the intermediate unfolding. Zero decay out to 2000 digits, radix 60, and 100% digit anonymization (exp80). This is the *generalize* side.

Decoupling (C4). The two facts — hard to obtain, easy to skip — are not in tension. Obtaining requires compiling the structure (training); generalizing uses the compiled structure (inference). The theory of P1 explains both: definition completeness + relational sufficiency determine whether compilation succeeds; step-skipping is what compiled structure does on new inputs.

4 Experiments

All via the official run_exp judge path (full-sequence reconstruction). Model: exp02_supervised_s2 (2-layer, dim-64, <1 MB, judgment 1.000 on the full suite).

4.1 Three-channel equivalence (C2)

Input	Construction	Formal	Intuition
12+34	✓	✓	✓
123456789+987654321	✓	✓	✓
999...9 (20 digits)+1	✓	✓	✓
999...9 (2000 digits)+1	✓	✓	1.000

Construction = formal exactly on full output sequences (2/9/20/2000 digits). Intuition scores 1.000 on the same inputs. The three channels realize the same computation.

4.2 Intuition carries structure, not symbols (C3)

- **Symbol permutation (exp41):** 26080 digit renamings during training, tested on original symbols — invariance 0.996→0.987. The intuition channel realizes structural relations (successor/iteration/carry), not digit signifiers.
- **Step-skipping zero decay (exp80):** 2000 digits / radix 60 / 100% anonymization / joint, all 1.000. The compiled structure extrapolates without decay.
- **Machine form (exp90):** 20-digit addition compiles 82 construction tokens into 5 intuition tokens — the skip is visible in the token stream. Honest caveat: the atomic token is outside the model vocab (pad id 0); measurement is the sequence-length effect, a lower bound.

4.3 Acquisition-generalization decoupling (C4)

Epochs	1	2	3	5	8
Judgment acc (full suite)	0.001	0.214	0.244	0.781	0.996

Acquisition is monotone but config-dependent (8 epochs for the full suite). The same converged model then extrapolates with zero decay (§4.2). The “hard to obtain” and “easy to skip” facts coexist; they describe obtain (training) and generalize (inference), respectively.

5 Discussion

What the equivalence does not say. We do not claim the three philosophical positions are identical in their epistemology; we claim they are equivalent as *execution channels of one system*. The historical disagreement concerned the nature of mathematical truth; our claim concerns the architecture of a computing system that realizes mathematical objects. The philosophical positions are models of what a mathematical object is; we show one system can realize all three models of the same object consistently.

Anti-Platonism. Intuition is compiled construction, not an a priori faculty. If mathematical structure were innate (Platonism / nativism), then definitionally-impooverished concepts would still generalize; they do not (P1 §8: an impoverished vocabulary requires an order of magnitude more training; genuinely under-defined tokens never generalize). The intuition channel’s competence is earned by compilation, not endowed.

Positional vs constructive content. Structuralism says an object is its position; the intuition channel locates the object in the learned structure. Intuitionism says an object is its construction; the construction channel unfolds it. The equivalence of the channels shows the two descriptions coincide — for objects whose definitions are complete. This is the machine-level form of the old structuralist-intuitionist rapprochement.

6 Related Work

- **Computational Trinitarianism** (Curry-Howard): programs = proofs = types. We extend the three-way identification to execution channels of one neural system.
- **Constructive mathematics / type theory** (Martin-Löf 1984; Bishop): existence = constructibility. Our construction channel is its machine form.
- **Structuralism / Bourbaki; Gärdenfors conceptual spaces**: object = position in a structure/geometry. Our intuition channel is its neural realization.
- **Formalism / Hilbert; rewriting systems**: object = sign manipulation. Our formal channel is the definition-driven engine.
- **Dual-process cognition** (Kahneman): System 1/2. Our intuition/reasoning separation is the System 1/2 split with a compilation account of how System 1 is built.
- **Dual-systems in math cognition** (e.g. number sense / deliberate arithmetic): our three channels refine the two-system picture with a verifiable equivalence.

6.5 Mathematical-domain validation: the Riemann-direction case

The three-channel account is not confined to arithmetic. In a companion formalization (Lean 4 / mathlib, claims C011-C025, all PROVED, novelty KNOWN, no sorry), the intuition chain “the complex axis is a high-dimensional projection of -1 ; primes land on translated integer points; $1/2$ is the inversion-translation symmetry center; the complex axis is curled” was carried through all three channels:

- **Construction** (definition-driven): the ComplexAxis algebra ($a + bj$, $j^2 = -1$), projection $\text{proj}\langle a, b \rangle = a$, lifting, basepoint drift;

- **Formal** (rewriting): Lean 4 proofs of projection non-preservation, prime-circle structure (8 lattice points as a single orbit under rotation $\times$ conjugation, via Gaussian-integer UFD), Euler-product convergence ($\text{Re}(s) > 1$), and the critical-line geometry;
- **Intuition** (compiled structure): the same objects are reached directly by the intuition chain, skipping textbook derivation.

Projection recovery theorem (`proj_not_recoverable` / `proj_recoverable_symmetry`, Lean-proved): *lost structure is not recoverable; symmetry directions are recoverable*. Under projection, the imaginary-axis direction (i vs $-i$) is lost and cannot be recovered, while the real-axis sign symmetry (r vs $-r$) is preserved. This is the mathematical counterpart of EXP-61h (§4): the intuition channel is position-insensitive (it drops positional detail like projection drops the J direction) yet preserves semantic symmetry — matching “scrambled letters do not impede understanding; deliberate logic on the scramble loses meaning” (G5b, P1). Projection is the mechanism of the fast path: dropping irrelevant structure irrecoverably is what makes the direct route cheap.

Token economy of intuition-guided formalization (methodological data): the full C011-C025 formalization consumed $\approx 700k$ tokens, of which 99.2% were context-cache re-transmission; net new content $< 1\%$. The intuition chain hit known structures directly (heap/torsor, sums of two squares, circle inversion, functional-equation symmetry), avoiding textbook derivation — an efficiency datum for G5 at the token level, independent of the arithmetic experiments in §4.

7 Limitations

1. The equivalence is demonstrated on the supported calculus (formalized definitions), not on all of mathematics.
2. The intuition channel is statistical, not guaranteed; equivalence is verified (held-out) and falls back to construction, not proven.
3. The atomization result (exp90) is a sequence-length lower bound, not a claim of learned atomic semantics.

8 Conclusion

The three positions of the foundations dispute — construction, structure, sign — are realized as three equivalent execution channels of one neural system. Construction = formal exactly; intuition agrees at 1.000; the intuition channel carries structure, not symbols; and it is hard to obtain but easy to generalize by skipping steps. Intuition is compiled construction. A century-old split in the foundations of mathematics is, at least for a formalized calculus, an architectural fact about how many ways one system can realize the same object — and all of them agree.

Appendix

- A. Full experiment tables: `docs/paper_data/results.csv`.
- B. Relation to P1 propositions (G0, G5, R) and P2 compilation.
- C. Formal-verification direction (Lean): $\text{construction} \equiv \text{intuition} \equiv \text{formal}$, for the supported calculus.

Unified Conclusion (added 2026-08-11)

Once, under a unified formal attention-syntactic structure, a neural network ultimately, inevitably, and necessarily clearly, intuitively, and interpretably observes the boundary between

intuition and construction, we can with full confidence use the unified attention-formal language to direct logical construction, use precise synthetic-CoT logical construction to earn *flashing intuition*, and use distilled intuitive context to exhaustively search attention forms. Form, construction, and intuition co-adapt within the unified framework, forming a feedback iteration pipeline of teaching, training, and reasoning in which they are no longer distinct — perhaps one of the faster shortcuts among the many paths toward ASI.

Cross-References within the Unified Framework

This paper is one leg of the *Unified Framework of Representation, Logic, and Intuition* (manifesto: `Unified_Framework_Representation_Logic_Intuition.md`). The three legs co-adapt:

Leg	Paper	Claim
form → construction	P1 (this paper)	correct syntax → extreme training speed; correct construction → extreme generalization
construction → intuition	P0 (three-channel equivalence)	construction = formal = intuition; intuition is compiled construction
construction → intuition (method)	P2 (neural macro compilation)	compile the slow construction path into the fast reasoning path

Cyclic closure: form directs construction (P1), construction earns intuition (P0/P2), intuition searches form (the fast path extrapolates and re-expresses in the unified form). The unified conclusion — form directs logical construction, construction earns flashing intuition, intuition exhaustively searches forms — is stated in the manifesto and in each paper’s Unified Conclusion.